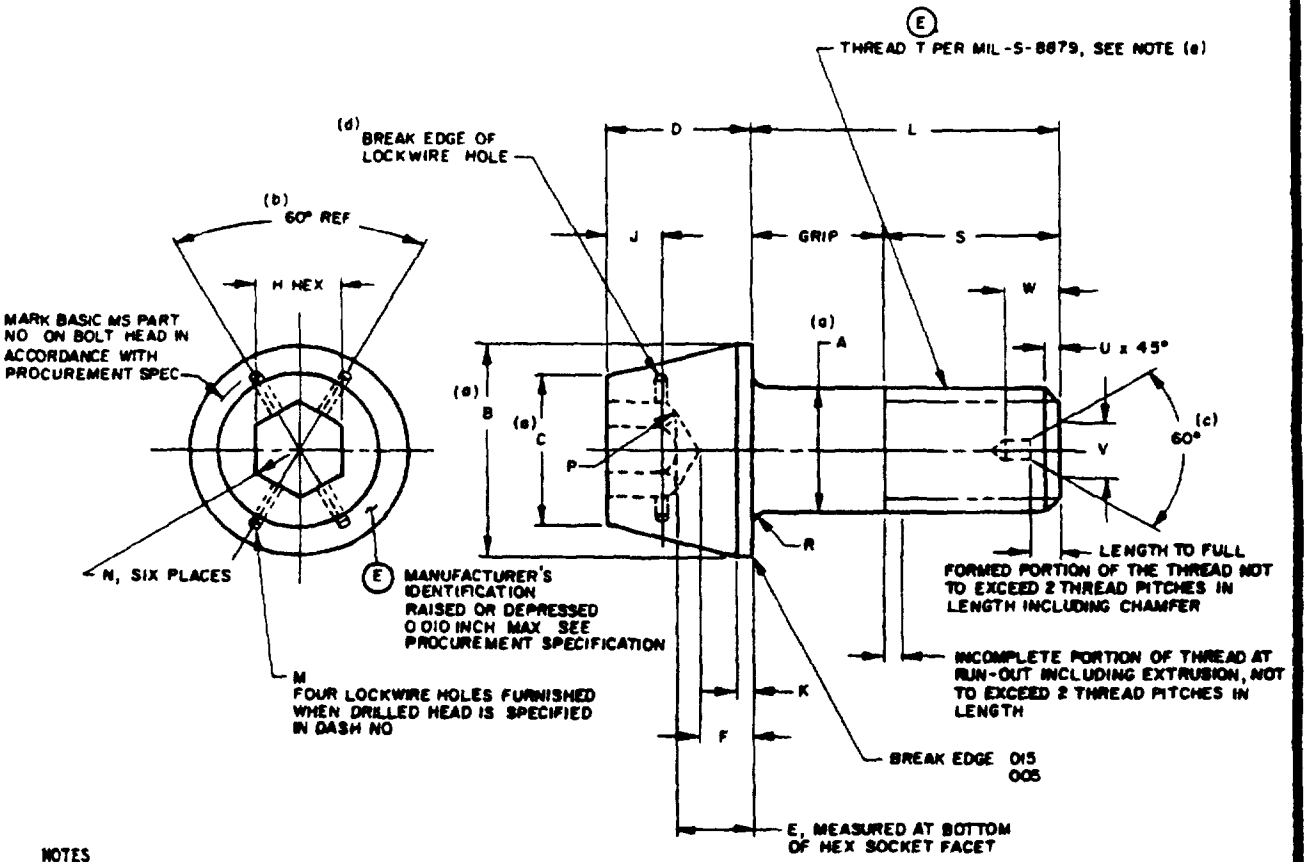
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NOTES

(a) DIMENSIONS A, B AND C SHALL BE CONCENTRIC TO EACH OTHER WITHIN VALUES SPECIFIED FOR X TOTAL INDICATOR READING DIMENSIONS SHALL BE H AND C SHALL BE CONCENTRIC TO EACH OTHER WITHIN VALUES SPECIFIED FOR Y TOTAL INDICATOR READING. DIMENSION A AND THREAD PITCH DIAMETER SHALL BE CONCENTRIC TO EACH OTHER WITHIN VALUES SPECIFIED FOR Z TOTAL INDICATOR READING.
(b) REFERENCE DIMENSIONS ARE FOR DESIGN PURPOSES AND TO PROVIDE INFORMATION ONLY. THESE DIMENSIONS ARE NOT INSPECTION OR MANUFACTURING REQUIREMENTS. S MIN REF = (L MIN) - (GRIP MAX).
(c) ON MS20012 THRU MS20024 CENTER DRILLING OF SHANK IS OPTIONAL FOR BOLT MANUFACTURER.
(d) COUNTERSINKING OF DRILLED HOLES IS OPTIONAL.
(E) (e) MS20016 (SEE TABLE I), 1-14NS-3A IS INACTIVE FOR NEW DESIGN MS20016 THREADS ARE NOTED AS AN EXCEPTION IN MIL-B-7838
(E) (f) MS20012A IS CANCELLED AFTER 13 JANUARY 1990

REQUIREMENTS

(E) 1 MATERIAL ALLOY STEEL - UNS G43400, UNS G43406, UNS G41400, UNS G87400, UNS G87350
2 FINISH CADMIUM PLATING PER QQ-P-416, TYPE II, CLASS 2 PARTS WITH CLASS 3 PLATING MAY BE FURNISHED FROM SUPPLIER'S STOCK UNTIL 1 JANUARY 1975.
(E) 3 HARDNESS 34-40 HRC
4. DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED, TOLERANCES DECIMALS - .010, ANGLES ±1°
5 DIMENSIONS TO BE MET AFTER PLATING
INTERCHANGEABILITY RELATIONSHIP BOLTS WITH TYPE II PLATING CAN BE USED TO REPLACE BOLTS WITH TYPE I PLATING IN ALL APPLICATIONS
BOLTS WITH TYPE I PLATING CAN BE USED IN PLACE OF BOLTS WITH TYPE II PLATING UNTIL PRESENT STOCKS ARE DEPLETED, BUT BOLTS WITH TYPE I PLATING SHALL NOT BE USED AFTER 1 JANUARY 1964

FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID

(E) DENOTES CHANGES

PREPARING ACTIVITY NAVY - AS CUSTODIANS ARMY-AR NAVY- AIR FORCE-99 DLA- REVIEW ARMY - ER, ME, MI DLA - IS AF-82 USER ARMY - AV PROJECT NUMBER 5306-1238	MILITARY SPECIFICATION SHEET TITLE BOLTS, INTERNAL WRENCHING, 160 KSI FTU AND 95 KSI FSU	SPECIFICATION SHEET NUMBER 30 MAY 1989 MS20004E THRU 20024E SUPERSEDING MS20004D THRU 20024D 10 AUGUST 1981 AMSC- N/A FSC 5306

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TABLE I BOLT DIMENSIONS AND TENSILE STRENGTH

BASIC MS PART NO	(a) THREAD T	(a) A DIA MAX	(a) A DIA MIN	(a) B DIA MAX	(a) B DIA MIN	(a) C DIA	D	E	F MIN	(a) H HEX MAX	(a) H HEX MIN	J	K	M +.005 - 000 DIA	N MAX RAD
MS20004	1/4-28 UNJF-3A	.2492	.2477	.438	.428	.342	.250	.112	.068	.1900	.1880	.078		.037	.010
MS20005	5/16-24 UNJF-3A	.3117	.3102	.531	.521	.394	.312	.144	.094	.2210	.2190	.094			
MS20006	3/8-24 UNJF-3A	.3742	.3727	.649	.639	.520	.375	.183	.116	.3150	.3130	.125	.063		.011
MS20007	7/16-20 UNJF-3A	.4367	.4347	.750	.740	.534	.438	.208	.141			.156			
MS20008	1/2-20 UNJF-3A	.4991	.4971	.828	.818	.625	.500	.247	.169	.3785	.3755	.172			
MS20009	9/16-18 UNJF-3A	.5616	.5596	.938	.928	.723	.562	.279	.190	.4410	.4380	.203			
MS20010	5/8-18 UNJF-3A	.6240	.6220	1.050	1.040	.826	.625	.317	.216	.5035	.5005	.234			.019
MS20012	3/4-16 UNJF-3A	.7488	.7468	1.230	1.220	.921	.750	.385	.273	.5660	.5630	.281			
MS20014	7/8-14 UNJF-3A	.8737	.8707	1.438	1.428	1.083	.875	.450	.327	.6295	.6257	.344	.094	.055	
MS20016 *	1 -14 NS -3A	.9985	.9955	1.625	1.615	1.249	1.000	.527	.381	.7547	.7507	.391			.022
MS20017	1 -12 UNJF-3A														
MS20018	1-1/8-12 UNJF-3A	1.124	1.121	1.875	1.865	1.275	1.125	.589	.443			.453			
MS20020	1-1/4-12 UNJF-3A	1.249	1.246	2.125	2.115	1.598	1.250	.667	.475	1.005	1.001	.500	.125		
MS20022	1-3/8-12 UNJF-3A	1.374	1.370	2.313	2.303	1.768	1.375	.746	.531	1.131	1.126	.547			
MS20024	1-1/2-12 UNJF-3A	1.499	1.495	2.500	2.490	1.931	1.500	.823	.586	1.256	1.251	.594	.156		
~~MS20012A~~	~~3/4-16 UNJF-3A~~	~~.7488~~	~~.7468~~	~~1.230~~	~~1.220~~	~~.921~~	~~.750~~	~~.385~~	~~.273~~	~~.5660~~	~~.5630~~	~~.281~~	~~.094~~		

(E) SEE NOTE (f)

TABLE I (CONTINUED)

BASIC MS PART NO	P RAD MAX	P RAD MIN	P RAD MAX	P RAD MIN	(b) S MIN REF	U ±.016	(c) V MAX	(c) W MAX	(b) (c) CENTER DRILL SIZE REF	TENSILE STRENGTH LB MIN	(a) CONCENTRICITY X	Y	Z
MS20004	020	010	.041	.031	.475	.031				6,190	.005	.007	0045
MS20005					.537					9,820	.006	.008	
MS20006					.662	.047				15,200	.008	.010	
MS20007			.057	.047	.787					20,600	.009	.011	.006
MS20008										27,400	.010	.012	
MS20009					.850					34,800	.011	.014	
MS20010					.912	.062				43,600	.012	.016	
MS20012	030	015	073	063	1.037		187	200	2	63,200	015	.018	
MS20014					1.162	.078				86,100	018	.022	
MS20016 *					1.287		250	260	3	112,000	.020	.025	.009
MS20017										114,000			
MS20018					1.475	.094				144,000	.022	.026	
MS20020			089	.077	1.600		.312	320	4	180,000	.025	.032	
MS20022					1.725					219,000	028	.035	.012
MS20024					1.850					263,000	.030	.039	
~~MS20012A~~			~~073~~	~~063~~	~~0.730~~	~~062~~	~~187~~	~~200~~	~~2~~	~~63,200~~	~~015~~	~~018~~	~~006~~

(E) SEE NOTE (f)

* SEE NOTE (*)

PREPARING ACTIVITY NAVY - AS CUSTODIANS ARMY - AR NAVY - AIR FORCE - 99 DLA - ARMY - EP, ME, MI DLA - IS AF-HC REVIEW ACTIVITIES USER ARMY - AV PROJECT NUMBER 5306-1233	MILITARY SPECIFICATION SHEET TITLE BOLTS, INTERNAL WRENCHING, 160 KSI FTU AND 96 KSI FSU	SPECIFICATION SHEET NUMBER 30 MAY 1989 MS20004E THRU 20024E SUPERSEDING MS20004D THRU 20024D 10 AUGUST 1981 AMSC - N/A FSC 5306

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TABLE II DASH NOS, GRIP LENGTHS AND BOLT LENGTH

DASH NO	GRIP ± 010	LENGTH ± 015 MS20004 1/4	MS20005 5/16	MS20006 3/8	MS20007 7/16	MS20008 1/2	MS20009 9/16	MS20010 5/8	MS20012 3/4	MS20014 7/8	MS20016 1	MS20017 1	MS20018 1-1/8	MS20020 1-1/4	MS20022 1-3/8	MS20024 1-1/2
4	.250	.750														
6	375	875	938	1 062												
8	500	1.000	1 062	1 188	1 312	1 312										
10	625	1.125	1 188	1.312	1 438	1 438	1.500	1.562	*1 856							
12	750	1 250	1.312	1 438	1.562	1 562	1 625	1.688	1 812							
14	.875	1.375	1 438	1 562	1 688	1 688	1 750	1 812	1 938	2 062						
16	1 000	1 500	1.562	1 688	1 812	1 812	1.875	1 938	2 062	2 188	2 312	2 312				
18	1.125	1.625	1.688	1 812	1 938	1 938	2 000	2 062	2 188	2 312	2 438	2.438	2 625			
20	1.250	1 750	1 812	1 938	2 062	2 062	2 125	2 188	2 312	2 438	2.562	2 562	2 750	2 875		
22	1 375	1 875	1 938	2 062	2 188	2 188	2 250	2 312	2 438	2 562	2.688	2 688	2 875	3.000	3 125	
24	1 500	2 000	2.062	2 188	2.312	2 312	2 375	2 438	2.562	2.688	2 812	2.812	3.000	3 125	3 250	3.375
26	1.625	2.125	2.188	2.312	2 438	2 438	2.500	2.562	2 688	2 812	2 938	2 938	3 125	3 250	3 375	3.500
28	1 750	2.250	2 312	2 438	2.562	2 562	2 625	2 688	2 812	2 938	3 062	3 062	3 250	3.375	3 500	3 625
30	1.875	2 375	2.438	2 562	2 688	2 688	2 750	2.812	2 938	3 062	3 188	3 188	3.375	3 500	3 625	3.750
32	2.000	2 500	2 562	2 688	2 812	2 812	2 875	2 938	3.062	3 188	3 312	3.312	3 500	3 675	3 750	3.875
34	2 125	2 625	2.688	2 812	2 938	2 938	3 000	3 062	3 188	3 312	3 438	3 438	3.625	3 750	3.875	4 000
36	2 250	2 750	2 812	2 938	3 062	3 062	3 125	3 188	3 312	3 438	3 562	3.562	3 750	3 875	4 000	4.125
38	2.375	2.875	2.938	3.062	3 188	3.188	3.250	3 312	3 438	3 562	3 688	3 688	3 875	4 000	4 125	4.250
40	2 500	3 000	3 062	3 188	3 312	3.312	3.375	3 438	3 562	3 688	3 812	3 812	4 000	4 125	4.250	4.375
42	2 625	3 125	3 188	3.312	3 438	3 438	3 500	3 562	3 688	3 812	3.938	3 938	4 125	4 250	4 375	4 500
44	2 750	3 250	3 312	3 438	3 562	3 562	3 625	3 688	3 812	3 938	4 062	4 062	4 250	4 375	4 500	4.625
46	2 875	3 375	3.438	3 562	3 688	3.688	3.750	3 812	3 938	4 062	4 188	4.188	4.375	4 500	4 625	4.750
48	3 000	3 500	3 562	3 688	3 812	3.812	3 875	3 938	4 062	4 188	4 312	4 312	4.500	4 625	4 750	4 875
50	3 125	3.625	3 688	3 812	3 938	3.938	4.000	4 062	4 188	4.312	4 438	4 438	4 625	4 750	4 875	5.000
52	3 250	3 750	3 812	3 938	4.062	4 062	4 125	4.188	4.312	4 438	4 562	4 562	4 750	4 875	5 000	5.125
54	3.375	3 875	3.938	4.062	4 188	4 188	4 250	4 312	4 438	4 562	4.688	4 688	4 875	5 000	5 125	5 250
56	3 500	4 000	4 062	4 188	4 312	4 312	4 375	4 438	4 562	4 688	4.812	4.812	5 000	5 125	5 250	5.375
58	3 625	4.125	4 188	4 312	4 438	4 438	4 500	4.562	4 688	4 812	4.938	4 938	5 125	5 250	5.375	5.500
60	3.750	4.250	4 312	4 438	4 562	4 562	4 625	4.688	4 812	4.938	5.062	5 062	5 250	5.375	5.500	5.625
62	3 875	4.375	4 438	4 562	4 688	4 688	4 750	4 812	4.938	5 062	5 188	5 188	5 375	5 500	5.625	5.750
64	4 000	4.500	4 562	4 688	4.812	4 812	4.875	4 938	5 062	5 188	5 312	5 312	5 500	5 625	5 750	5.875
66	4.125	4.625	4.688	4.812	4.938	4 938	5 000	5.062	5 188	5.312	5 438	5.438	5 625	5 750	5 875	6 000
68	4 250	4 750	4.812	4 938	5 062	5 062	5.125	5.188	5.312	5 438	5 562	5 562	5 750	5 875	6 000	6.125
70	4 375	4 875	4 938	5.062	5 188	5 188	5 250	5 312	5 438	5 562	5.688	5 688	5.875	6 000	6 125	6.250
72	4 500	5 000	5 062	5 188	5 312	5 312	5 375	5 438	5.562	5 688	5.812	5 812	6.000	6 125	6 250	6 375
74	4.625	5.125	5.188	5.312	5 438	5.438	5.500	5.562	5.688	5.812	5.938	5 938	6 125	6 250	6.375	6.500
76	4 750	5 250	5 312	5 438	5.562	5 562	5 625	5 688	5 812	5 938	6 062	6 062	6.250	6 375	6 500	6.625
78	4.875	5.375	5.438	5.562	5.688	5.688	5 750	5 812	5.938	6 062	6 188	6 188	6 375	6.500	6 625	6.750
80	5 000	5 500	5 562	5 688	5 812	5 812	5.875	5 938	6.062	6 188	6 312	6 312	6 500	6 625	6 750	6.875
82	5 125	5 625	5.688	5 812	5.938	5 938	6.000	6.062	6 188	6 312	6 438	6 438	6.625	6 750	6 875	7.000
84	5 250	5 750	5.812	5 938	6 062	6 062	6.125	6 188	6 312	6 438	6 562	6 562	6 750	6.875	7.000	7 125
86	5.375	5 875	5 938	6 062	6 188	6 188	6.250	6.312	6 438	6 562	6.688	6 688	6 875	7 000	7.125	7.250
88	5 500	6 000	6 062	6 188	6 312	6 312	6 375	6 438	6 562	6 688	6 812	6 812	7.000	7 125	7 250	7.375
90	5.625	6 125	6 188	6 312	6 438	6 438	6 500	6 562	6 688	6 812	6 938	6.938	7.125	7.250	7.375	7 500
92	5 750	6.250	6 312	6 438	6 562	6 562	6 625	6.688	6 812	6 938	7.062	7 062	7.250	7 375	7.500	7.625
94	5 875	6.375	6 438	6 562	6 688	6.688	6 750	6.812	6 938	7 062	7.188	7 188	7 375	7.500	7.625	7 750
96	6 000	6 500	6 562	6 688	6 812	6 812	6.875	6 938	7 062	7 188	7 312	7 312	7 500	7 625	7 750	7 875
98	6.125						7 000	7 062	7 188	7 312	7 438	7 438	7 625	7 750	7 875	8 000
100	6 250						7 125	7 188	7 312	7 438	7 562	7 562	7 750	7 875	8 000	8 125
102	6.375						7 250	7 312	7 438	7 562	7.688	7.688	7 875	8 000	8 125	8 250
104	6 500						7 375	7 438	7 562	7 688	7 812	7 812	8 000	8 125	8 250	8 375
106	6 625						7 500	7 562	7 688	7 812	7 938	7.938	8 125	8.250	8 375	8 500
108	6 750						7.625	7 688	7 812	7.938	8 062	8.062	8 250	8 375	8 500	8 625
110	6 875						7 750	7 812	7 938	8 062	8 188	8 188	8 375	8.500	8 625	8 750
112	7.000						7.875	7 938	8 062	8 188	8 312	8 312	8.500	8 625	8 750	8.875
114	7 125												8 625	8 750	8 875	9.000
116	7 250												8 750	8 875	9 000	9 125
118	7 375												8 875	9 000	9 125	9.250
120	7 500												9 000	9 125	9 250	9 375
122	7 625												9 125	9 250	9 375	9 500
124	7 750												9 250	9 375	9 500	9 625
126	7 875												9 375	9 500	9.625	9 750
128	8 000												9 500	9 625	9 750	9 875

(E) *APPLIES TO MS20012A ONLY.

PREPARING ACTIVITY NAVY - AS
CUSTODIANS ARMY-AP NAVY-
AIR FORCE-99 DLA-
REVIEW ARMY - EP, ME, MI DLA - IS AF-8?
USER ARMY - AV
PROJECT NUMBER 5306-1238

MILITARY SPECIFICATION SHEET

TITLE BOLTS, INTERNAL WRENCHING, 160 KSI FTU AND 96 KSI FSU

SPECIFICATION SHEET NUMBER 30 MAY 1989
MS20004E THRU 20024E
SUPERSEDING
MS20004D THRU 20024D 10 AUGUST 1981
AMSC- N/A FSC 5306

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UNASSIGNED DASH NUMBERS SHALL NOT BE USED, EXCEPT THOSE DASH NUMBERS COVERED BY THE FOLLOWING CODING NOTE

DASH NUMBERS INDICATE PREFERRED GRIP LENGTHS IN 125 INCREMENTS INTERMEDIATE GRIP LENGTHS BETWEEN THOSE LENGTHS LISTED, MAY BE SPECIFIED IN .062 INCREMENTS BY INTERMEDIATE DASH NUMBERS LENGTH AND GRIP OF A BOLT WITH AN ODD NUMBER FOR THE DASH NUMBER WILL BE .062 GREATER THAN SHOWN FOR THE BOLTS WITH THE PRECEDING EVEN DASH NUMBER ONLY BOLT DIAMETER FOR WHICH THERE ARE QUALIFIED PRODUCTS LISTED ON QPL-7838 SHALL BE USED FOR DESIGN

FOR EASH BOLT DIAMETER GRIP LENGTHS GREATER OR LESS THAN THOSE DESIGNATED BY ASSIGNED DASH NUMBERS ON TABLE II ARE INACTIVE FOR DESIGN

ADD "H" BEFORE DASH NUMBER FOR BOLT WITH DRILLED HEAD

EXAMPLES OF PART NUMBERS MS20017-18 = 1-12 BOLT, 2.438 INCH LONG, 1.125 INCH GRIP, UNDRILLED HEAD
MS20017H18 = 1-12 BOLT, 2.438 INCH LONG, 1.125 INCH GRIP, DRILLED HEAD.

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